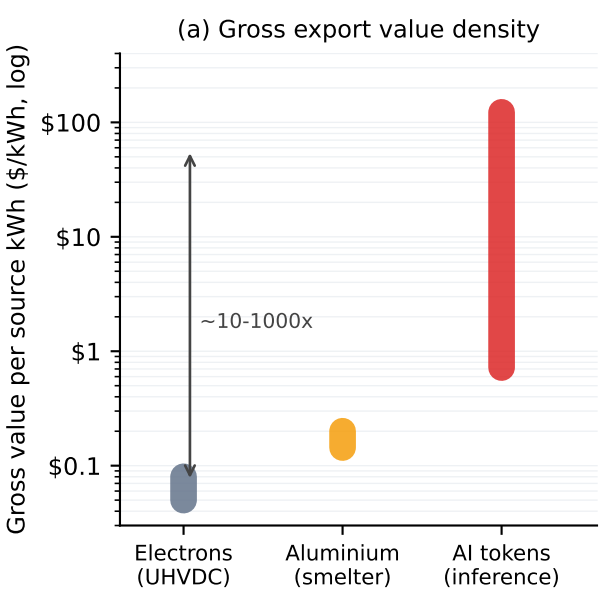
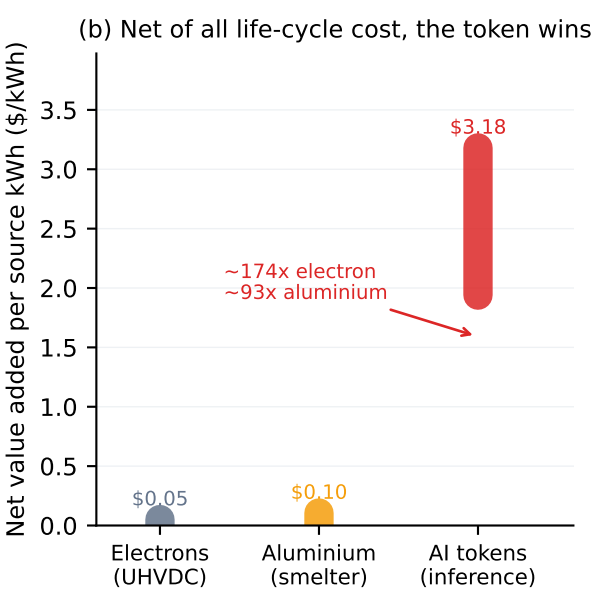
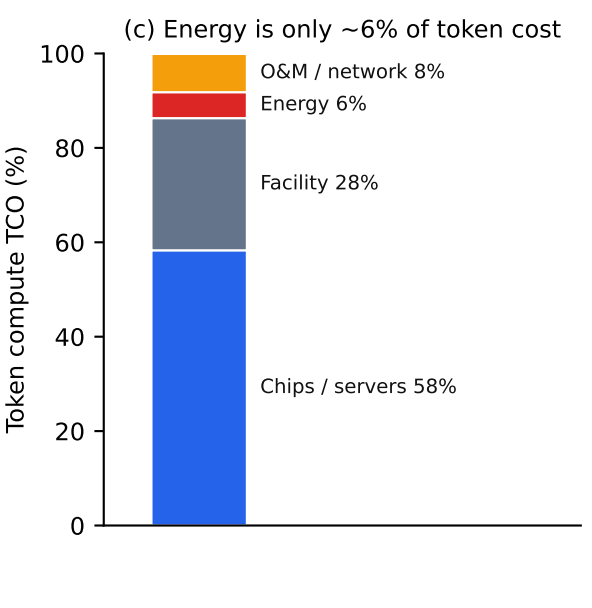

(a) Gross export value density
Gross value per source kWh ($/kWh, log)
$100
$10
$1
$0.1
~10-1000x
Electrons (UHVDC)
Aluminium (smelter)
AI tokens (inference)


(b) Net of all life-cycle cost, the token wins
Net value added per source kWh ($/kWh)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
$3.18
~174x electron
~93x aluminium
$0.05
$0.10
Electrons (UHVDC)
Aluminium (smelter)
AI tokens (inference)


(c) Energy is only ~6% of token cost
Token compute TCO (%)
100
80
60
40
20
0
O&M / network 8%
Energy 6%
Facility 28%
Chips / servers 58%